COMPUTER NETWORKS LAB: VIVA PREPARATION CHEAT SHEET

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Labs 6-10 Quick Reference

LAB 6: CONCURRENT SOCKET PROGRAMMING IN TCP

Key Concepts

Concurrent Server: Handles multiple client requests simultaneously using fork()

- Parent process continues accepting new connections
- Child process handles individual client communication

Fork() Function

- Creates exact copy of calling process
- Returns 0 in child process
- Returns child PID in parent process
- Returns -1 on error

Important System Calls

```
pid_t pid = fork();
if(pid == 0) {
    // Child process code
    // Handle client communication
    close(newsockfd);
    exit(0);
} else {
    // Parent process code
    close(newsockfd);
}
```

Key Points for Viva

- **Iterative vs Concurrent**: Iterative serves one client at a time; Concurrent serves multiple clients simultaneously
- **Why use fork()**: To create separate process for each client
- **getpid()**: Returns process ID
- **getppid()**: Returns parent process ID
- **wait()**: Parent waits for child process to finish

Common Lab Exercises

1. Concurrent echo server
2. Concurrent remote math server (arithmetic operations)
3. Concurrent duplicate word removal server
4. Concurrent daytime server with process ID

LAB 7: SOCKET PROGRAMMING APPLICATIONS

Objectives

Practical applications of socket programming for real-world scenarios

Typical Applications

1. **Encryption/Decryption System**
   - Client encrypts message (e.g., ASCII + 4)
   - Server decrypts and displays both forms
2. **Chat Application**
   - Multiple clients connect to server
   - Server broadcasts messages to all clients
   - Real-time communication
3. **Remote Math Server**
   - Client sends two numbers and operator
   - Server performs calculation
   - Returns result to client

Key Concepts

- **Concurrent connections**: Multiple clients simultaneously
- **Broadcasting**: Sending data to all connected clients
- **State management**: Tracking multiple client connections

Viva Questions to Prepare

- How to handle multiple clients?
- What is broadcasting in socket programming?
- How to maintain list of connected clients?
- Difference between TCP and UDP for chat applications?

LAB 8: ADVANCED IP CONFIGURATION - SUBNETTING & STATIC ROUTING

IP Address Classes

Class	Range	Default Mask	Network Bits	Host Bits
A	1-126	255.0.0.0 (/8)	8	24
B	128-191	255.255.0.0 (/16)	16	16
C	192-223	255.255.255.0 (/24)	24	8
D	224-239	Multicast	-	-
E	240-255	Experimental	-	-

Subnetting Formulas

- **Number of subnets**: 2^n (where n = borrowed bits)
- **Number of hosts per subnet**: 2^h - 2 (where h = host bits)
    - Subtract 2 for network address and broadcast address

Subnetting Steps

1. Determine number of required subnets/hosts
2. Calculate bits needed: 2^n ≥ required number
3. New subnet mask = original + borrowed bits
4. Calculate subnet ranges

Example: 192.168.1.0/24 into 4 subnets

- Need 2 bits (2^2 = 4)
- New mask: /26 (255.255.255.192)
- Hosts per subnet: 2^6 - 2 = 62
- Subnets:
    - 192.168.1.0/26 (0-63)
    - 192.168.1.64/26 (64-127)

- 192.168.1.128/26 (128-191)
- 192.168.1.192/26 (192-255)

Static Routing

Command: `ip route <destination_network> <subnet_mask> <next_hop_address>`

Example:

```
Router(config)# ip route 192.168.2.0 255.255.255.0 11.0.0.2
```

Key Points

- Static routes: Manual configuration, suitable for small networks
- Routing table: Stores network paths
- Next-hop address: IP of next router in path
- Administrative distance: Trustworthiness (Static = 1)

Viva Questions

- What is subnetting and why use it?
- Calculate subnets for given network
- Difference between static and dynamic routing?
- What is CIDR notation?
- What is default gateway?

LAB 9: VLAN CONFIGURATION & INTER-VLAN COMMUNICATION

VLANs (Virtual LANs)

Definition: Logical segmentation of physical network into separate broadcast domains

Benefits

- **Security**: Isolate sensitive data
- **Performance**: Reduce broadcast traffic
- **Management**: Easier network organization
- **Flexibility**: Group devices logically, not physically

Switch Port Modes

1. **Access Port**
    - Belongs to single VLAN
    - Connects end devices (PCs, printers)
    - Untagged traffic
2. **Trunk Port**
    - Carries traffic for multiple VLANs
    - Connects switches or switch-to-router

    - Uses 802.1Q tagging

VLAN Configuration Commands

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10

Switch(config)# interface fa0/24
Switch(config-if)# switchport mode trunk
```

Inter-VLAN Routing: Router-on-a-Stick

Concept: Single router interface with multiple sub-interfaces (one per VLAN)

Configuration:

```
Router(config)# interface g0/0
Router(config-if)# no shutdown

Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Key Components

- **Physical Interface**: Must be enabled (no shutdown)
- **Sub-interfaces**: Logical interfaces for each VLAN
- **802.1Q Encapsulation**: VLAN tagging protocol
- **Trunk Port**: On switch connected to router

SVI (Switch Virtual Interface)

- Logical interface on Layer 3 switch
- Provides routing services for VLAN
- One SVI per VLAN

Viva Questions

- What is VLAN and why use it?
- Difference between access and trunk port?
- Why does ping fail between different VLANs initially?
- What is 802.1Q tagging?
- Explain Router-on-a-Stick configuration
- What is native VLAN?

LAB 10: DYNAMIC ROUTING WITH RIP & DHCP/DNS CONFIGURATION

RIP (Routing Information Protocol)

Type: Distance-vector dynamic routing protocol

Characteristics:

- **Metric**: Hop count (max 15 hops)
- **Version 1**: Classful, no subnet info, broadcast updates
- **Version 2**: Classless, supports VLSM, multicast updates
- **Updates**: Every 30 seconds
- **Algorithm**: Bellman-Ford

RIP Configuration

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.10.0
Router(config-router)# network 192.168.20.0
Router(config-router)# no auto-summary
```

Static vs Dynamic Routing

Feature	Static	Dynamic (RIP)
Configuration	Manual	Automatic
Scalability	Low	High
Overhead	None	High
Convergence	N/A	Slow (RIP)
Use Case	Small networks	Medium-large networks

DHCP (Dynamic Host Configuration Protocol)

Purpose: Automatically assign IP addresses to devices

DHCP Process (DORA):

1. **Discover**: Client broadcasts request
2. **Offer**: Server offers IP address
3. **Request**: Client requests offered IP
4. **Acknowledge**: Server confirms assignment

DHCP Configuration on Router

```
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)# ip dhcp pool POOL-NAME
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 192.168.10.100
```

DHCP Relay Agent

Purpose: Forward DHCP requests across subnets

```
Router(config)# interface g0/0
Router(config-if)# ip helper-address 192.168.30.100
```

DNS (Domain Name System)

Purpose: Resolve domain names to IP addresses

DNS Configuration:

```
Router(config)# ip dns server
Router(config)# ip host www.example.com 192.168.10.100
```

Testing DNS:

```
PC&gt; nslookup www.example.com
```

Key Concepts

- **Routing Table**: Automatically populated by RIP
- **Convergence**: Time for all routers to have consistent routing info
- **Hop Count**: Number of routers between source and destination
- **IP Pool**: Range of addresses DHCP can assign
- **DNS A Record**: Maps domain name to IP address

Viva Questions

- What is RIP? Explain versions
- How does RIP calculate best path?
- What is hop count limit in RIP?
- Explain DHCP DORA process
- What is DHCP relay agent and when is it needed?
- Difference between DNS and DHCP?
- What is excluded-address in DHCP?
- Why use dynamic routing over static?

COMMON ROUTER/SWITCH MODES

Router/Switch CLI Modes

1. **User EXEC Mode**: `Router>`
   - Basic monitoring only
2. **Privileged EXEC Mode**: `Router#`
   - Command: `enable`
   - Access all commands

3. **Global Configuration Mode**: `Router(config)#`
   - Command: `configure terminal`
   - Device-wide configuration
4. **Interface Configuration Mode**: `Router(config-if)#`
   - Command: `interface <type><number>`
   - Configure specific interface
5. **Sub-interface Mode**: `Router(config-subif)#`
   - Command: `interface g0/0.10`
   - Configure sub-interface
6. **Router Configuration Mode**: `Router(config-router)#`
   - Command: `router rip`
   - Configure routing protocol

ESSENTIAL COMMANDS REFERENCE

Router Configuration

```
Router# enable
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# copy running-config startup-config
```

Testing Connectivity

```
ping &lt;ip-address&gt;
traceroute &lt;ip-address&gt;
show ip route
show ip interface brief
show running-config
```

VLAN Commands

```
Switch(config)# vlan 10
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# switchport mode trunk
```

Show Commands

```
show vlan brief
show interfaces trunk
show ip route
show ip dhcp binding
```

```
show ip dhcp pool
show running-config
```

KEY NETWORKING CONCEPTS

OSI Model Layers (Focus on these)

- **Layer 7 - Application**: HTTP, FTP, DNS, DHCP
- **Layer 4 - Transport**: TCP, UDP
- **Layer 3 - Network**: IP, Routing, ICMP
- **Layer 2 - Data Link**: MAC, Switching, VLANs
- **Layer 1 - Physical**: Cables, signals

Port Numbers to Remember

- HTTP: 80
- HTTPS: 443
- FTP: 20, 21
- DNS: 53
- DHCP: 67 (server), 68 (client)
- Telnet: 23
- SSH: 22

Important Protocols

- **TCP**: Connection-oriented, reliable, ordered delivery
- **UDP**: Connectionless, unreliable, faster
- **ICMP**: Ping, error reporting
- **ARP**: Maps IP to MAC address

TROUBLESHOOTING TIPS

Cannot Ping Between Devices

1. Check IP configuration (ip address, subnet mask, gateway)
2. Verify physical/logical connectivity
3. Check routing table (`show ip route`)
4. Verify VLAN configuration
5. Check firewall/ACLs

DHCP Not Working

1. Verify DHCP pool configuration
2. Check excluded addresses
3. Verify DHCP relay agent if different subnet
4. Check if DHCP service is enabled

5. Verify default gateway and DNS in pool

Inter-VLAN Communication Fails

1. Check Router-on-a-Stick configuration
2. Verify trunk port configuration
3. Check sub-interface encapsulation
4. Verify IP addresses on sub-interfaces
5. Check default gateway on PCs

RIP Not Working

1. Verify RIP is enabled on all routers
2. Check network statements
3. Verify version 2 is used
4. Check if auto-summary is disabled
5. Wait for convergence (30-60 seconds)

QUICK FORMULAS

Subnetting

- Subnets = 2^(borrowed bits)
- Hosts = 2^(host bits) - 2
- Subnet increment = 256 - last octet of subnet mask

Network Calculation

- Network address = IP AND Subnet Mask
- Broadcast = Network + (2^host_bits - 1)
- First host = Network + 1
- Last host = Broadcast - 1

PRACTICE QUESTIONS

Lab 6 - Concurrent Sockets

1. What is the difference between iterative and concurrent servers?
2. Explain fork() and its return values
3. Why do we close newsockfd in parent process?
4. What happens if we don't call wait() in parent?

Lab 7 - Socket Applications

1. How to implement broadcast messaging?
2. What data structures track multiple clients?
3. How to handle client disconnection?

Lab 8 - Subnetting

1. Subnet 192.168.10.0/24 for 6 subnets
2. How many hosts in 192.168.1.0/26?
3. What is the broadcast address of 192.168.5.64/27?

Lab 9 - VLANs

1. Why can't VLAN 10 ping VLAN 20 initially?
2. What is 802.1Q encapsulation?
3. Configure Router-on-a-Stick for 3 VLANs

Lab 10 - RIP & DHCP

1. What is the metric used by RIP?
2. Explain DHCP DORA process
3. When is DHCP relay agent needed?
4. Difference between RIP v1 and v2?

REMEMBER FOR VIVA

1. **Be clear on concepts**: Understand WHY, not just HOW
2. **Know command syntax**: Practice writing commands
3. **Understand packet flow**: How data moves through network
4. **Know when to use what**: Static vs Dynamic, TCP vs UDP, etc.
5. **Real-world applications**: Why these technologies matter
6. **Troubleshooting approach**: Systematic problem-solving
7. **Safety first**: Always save config, use correct IPs

FINAL TIPS

✅ Understand the PURPOSE of each lab
✅ Practice commands in Packet Tracer
✅ Draw network diagrams while explaining
✅ Explain step-by-step configuration
✅ Know the difference between similar concepts
✅ Be ready with examples
✅ Admit if you don't know, don't guess
✅ Relate to real-world scenarios when possible

GOOD LUCK WITH YOUR VIVA!